1

Chapter 3 Surfaces **2**

3.1 Surfaces 2

3.2 Tangents and Normals 34

3.3 Derivatives 53

3.4 Special Types of Surfaces 64

3.5 Compact Surfaces 76

Chapter 3 Surfaces

3.1 Surfaces

We will consider only surfaces in $\mathbb{R}^3$.

A surface is a subset of $\mathbb{R}^3$ which looks like $\mathbb{R}^2$ in the vicinity of any point.

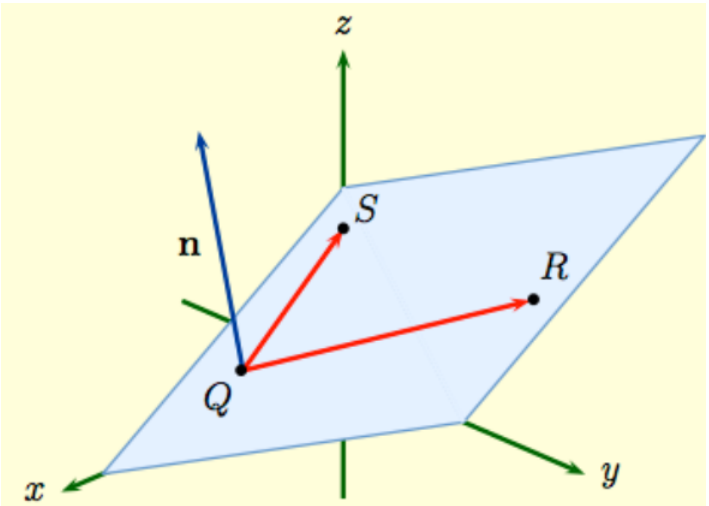
Let us first consider a plane in $\mathbb{R}^3$. A plane P in $\mathbb{R}^3$ is the set of points (x, y, z) satisfying an equation of the form

$$ax + by + cz = d,$$

where $\mathbf{n} = (a, b, c)$ is a nonzero vector. Namely,

$$P = \{(x, y, z) \,|\, ax + by + cz = d\}\,.$$

The vector $\mathbf{n}$ is a vector normal to the plane.



If S is a subset of $\mathbb{R}^3$, A is a subset of S, we say that A is open in S if there is an open set V of $\mathbb{R}^3$ such that

$$A = V \cap S.$$

Equivalently, for every $\mathbf{x}_0 \in S$, there exists $r > 0$ such that

$$B(\mathbf{x}_0, r) \cap S \subset A.$$

If S_1 and S_2 are subsets of $\mathbb{R}^3$, we say that a map $f : S_1 \to S_2$ is continuous if for every sequence $\{\mathbf{x}_k\}$ in S_1 that converges to $\mathbf{x}_0$ in S_1, the sequence $\{f(\mathbf{x}_k)\}$ converges to $f(\mathbf{x}_0)$ in S_2.

A map $f : S_1 \to S_2$ is continuous if and only if whenever B is an open subset of S_2,

$$f^{-1}(B) = \{\mathbf{x} \in S_1 \mid f(\mathbf{x}_1) \in B\}$$

is an open subset of S_1.

We say that $f : S_1 \to S_2$ is a homeomorphism if it is a continuous bijection and $f^{-1} : S_2 \to S_1$ is also continuous.

Given a plane

$$P = \{(x, y, z) \mid ax + by + cz = d\},$$

$\mathbf{n} = (a, b, c) \neq \mathbf{0}$. Assume that $c \neq 0$. Then the map $\boldsymbol{\sigma} : \mathbb{R}^2 \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}(u, v) = \left(u, v, \frac{d - au - bv}{c}\right)$$

is a *homeomorphism* between $\mathbb{R}^2$ and the plane.

The derivative matrix of $\boldsymbol{\sigma}$ is

$$\mathbf{D}\boldsymbol{\sigma}(u, v) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -\dfrac{a}{c} & -\dfrac{b}{c} \end{bmatrix}.$$

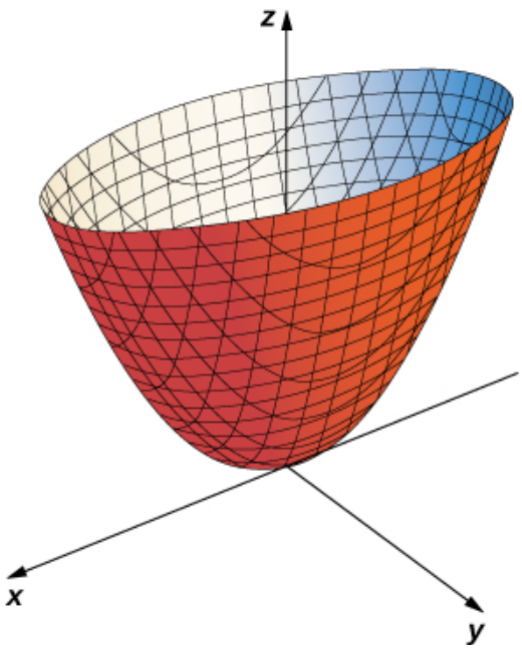
The two column vectors of $\mathbf{D}\boldsymbol{\sigma}(u, v)$, $\mathbf{w}_1 = (1, 0, -a/c)$ and $\mathbf{w}_2 = (0, 1, -b/c)$, are linearly independent. In fact, they are both perpendicular to the normal vector $\mathbf{a}$. Moreover,

$$\boldsymbol{\sigma}(u, v) = \mathbf{x}_0 + u\mathbf{w}_1 + v\mathbf{w}_2,$$

where $\mathbf{x}_0 = (0, 0, d/c)$ is a point on the plane.

More generally, let $\mathcal{O}$ be an open subset of $\mathbb{R}^2$, and let $f : \mathcal{O} \to \mathbb{R}$ be a smooth function. The graph of f is the subset of $\mathbb{R}^3$ given by

$$G_f = \{(x, y, z) \,|\, (x, y) \in \mathcal{O}, z = f(x, y)\}\,.$$



Let $\boldsymbol{\sigma} : \mathcal{O} \to \mathbb{R}^3$ be the mapping defined as

$$\boldsymbol{\sigma}(u, v) = (u, v, f(u, v)).$$

Then $\boldsymbol{\sigma} : \mathcal{O} \to \mathbb{R}^3$ is a smooth mapping, and it is a homeomorphism between $\mathcal{O}$ and G_f. At any $(u, v) \in \mathcal{O}$,

$$\mathbf{D}\boldsymbol{\sigma}(u, v) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ f_u(u, v) & f_v(u, v) \end{bmatrix}.$$

The two column vectors of $\mathbf{D}\boldsymbol{\sigma}(u, v)$ are linearly independent.

However, not all the surfaces of interest can be described as graphs of functions. The simplest example is the sphere $x^2 + y^2 + z^2 = 1$. Therefore, we need to generalize the definition of surfaces to include anything that locally looks like $\mathbb{R}^2$.

Definition 3.1 Regular Surfaces in $\mathbb{R}^3$

A subset S of $\mathbb{R}^3$ is a regular surface if for every point $\mathbf{x}_0$ in S, there is an open subset U of $\mathbb{R}^2$, an open subset V of $\mathbb{R}^3$ that contains $\mathbf{x}_0$, and a map $\boldsymbol{\sigma} : U \to V \cap S$,

$$\boldsymbol{\sigma}(u, v) = (\sigma_1(u, v), \sigma_2(u, v), \sigma_3(u, v))$$

that satisfies the following conditions:

(i) $\boldsymbol{\sigma} : U \to V \cap S$ is a bijection.

(ii) $\boldsymbol{\sigma} : U \to V \cap S$ is smooth.

(iii) For each $\mathbf{u}$ in U, the linear transformation $\mathbf{D}\boldsymbol{\sigma}(\mathbf{u}) : \mathbb{R}^2 \to \mathbb{R}^3$ is one-to-one.

The map $\boldsymbol{\sigma} : U \to V \cap S$ is a parametrization or a system of local coordinates of the open subset $V \cap S$ of S. Thus, $V \cap S$ is a coordinate neighbourhood of the point $\mathbf{x}_0$ in S.
The map $\boldsymbol{\sigma} : U \to V \cap S$ is also called a coordinate patch. A collection of coordinate patches whose images cover the whole S is called an atlas of S.

When there is no ambiguity, we will write a coordinate patch $\boldsymbol{\sigma} : U \to V \cap S$ simply as $\boldsymbol{\sigma} : U \to S$.

Notice that the columns of $\mathbf{D}\boldsymbol{\sigma}(\mathbf{u})$ are $\boldsymbol{\sigma}_u(u, v)$ and $\boldsymbol{\sigma}_v(u, v)$. The linear transformation defined by $\mathbf{D}\boldsymbol{\sigma}(u, v)$ is one-to-one if and only if the two vectors $\boldsymbol{\sigma}_u(u, v)$ and $\boldsymbol{\sigma}_v(u, v)$ are linearly independent, if and only if

$$\boldsymbol{\sigma}_u(u, v) \times \boldsymbol{\sigma}_v(u, v) \neq \mathbf{0}.$$

We also remark that under the conditions (i), (ii) and (iii), the map $\boldsymbol{\sigma}^{-1} : V \cap S \to U$ is also continuous. Hence, $\boldsymbol{\sigma} : U \to V \cap S$ is a homeomorphism.

Example 3.1

The plane $x + 2y + 3z = 6$ is a regular surface S with parametrization given by $\boldsymbol{\sigma} : \mathbb{R}^2 \to S$,

$$\boldsymbol{\sigma}(u, v) = (6 - 2u - 3v, u, v).$$

Example 3.2

Let $\mathcal{O}$ be an open subset of $\mathbb{R}^2$. Consider $\mathcal{O}$ as a subset of $\mathbb{R}^3$ via the natural inclusion, $\mathcal{O}$ is a regular surface.

Example 3.3

The paraboloid $z = x^2 + y^2$ is a regular surface S with parametrization given by $\boldsymbol{\sigma} : \mathbb{R}^2 \to S$,

$$\boldsymbol{\sigma}(u, v) = (u, v, u^2 + v^2).$$

Proposition 3.1

Let $\mathcal{O}$ be an open subset of $\mathbb{R}^2$, and let $f : \mathcal{O} \to \mathbb{R}$ be a smooth function. Then the graph of f, which is the subset of $\mathbb{R}^3$ given by

$$G_f = \{(x, y, z) \,|\, (x, y) \in \mathcal{O}, z = f(x, y)\},$$

is a regular surface.

Corollary 3.2

Let $\mathcal{O}$ be an open subset of $\mathbb{R}^2$, and let $f : \mathcal{O} \to \mathbb{R}$ be a smooth function. Then

(a) $\{(x, y, z) \mid (y, z) \in \mathcal{O}, x = f(y, z)\}$ is a regular surface.

(b) $\{(x, y, z) \mid (x, z) \in \mathcal{O}, y = f(x, z)\}$ is a regular surface.

Example 3.4

Consider the cylinder $C = \left\{(x, y, z) \,|\, x^2 + y^2 = 4\right\}$. Let

$$C_+ = C \setminus \{(-2, 0, z) \,|\, z \in \mathbb{R}\}\,, \quad C_- = C \setminus \{(2, 0, z) \,|\, y \in \mathbb{R}\}\,.$$

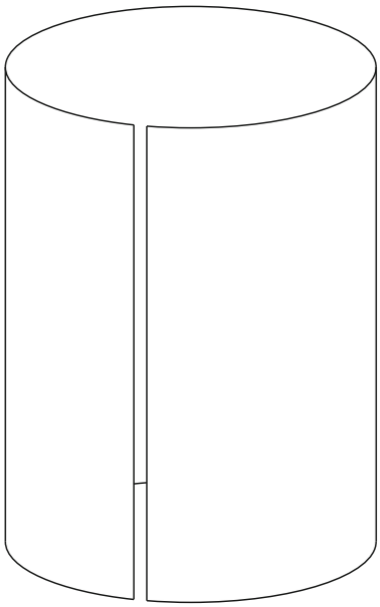
Notice that C_+ and C_- are open in C, and $C = C_+ \cup C_-$. The map $\boldsymbol{\sigma}_+ : (-\pi, \pi) \times \mathbb{R} \to C_+$,

$$\boldsymbol{\sigma}_+(u, v) = (2\cos u, 2\sin u, v)$$

is a parametrization of C_+; while the map $\boldsymbol{\sigma}_- : (0, 2\pi) \times \mathbb{R} \to C_-$,

$$\boldsymbol{\sigma}_-(u, v) = (2\cos u, 2\sin u, v)$$

is a parametrization of C_-. Hence, C is a regular surface with atlas $\{\boldsymbol{\sigma}_+, \boldsymbol{\sigma}_-\}$.



Proposition 3.3

If S is a regular surface, $\widetilde{S}$ is an open subset of S, then $\widetilde{S}$ is also a regular surface.

Example 3.5

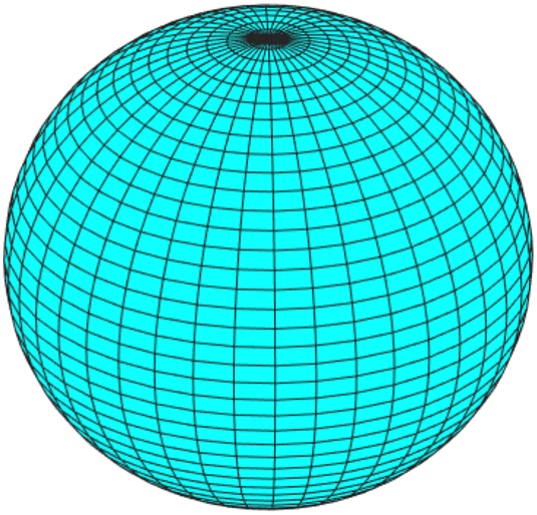
The set

$$S = \{(x, y, z) \mid x + 2y + 3z = 6, -1 < x < 2\}$$

is a regular surface.

Example 3.6

Show that the sphere $S = \left\{(x, y, z) \,|\, x^2 + y^2 + z^2 = 1\right\}$ is a regular surface.



The spherical coordinates of a point (x, y, z) in $\mathbb{R}^3$ is (ρ, ϕ, θ), where

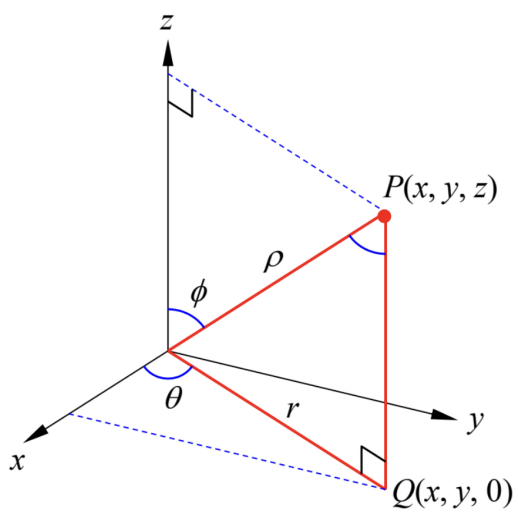
$$x = \rho \sin\phi \cos\theta, \quad y = \rho \sin\phi \sin\theta, \quad z = \rho \cos\phi.$$

The points on the sphere $x^2 + y^2 + z^2 = r^2$ of radius r can be parametrized by

$$(r \sin\phi \cos\theta, r \sin\phi \sin\theta, r \cos\phi),$$

where

$$0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

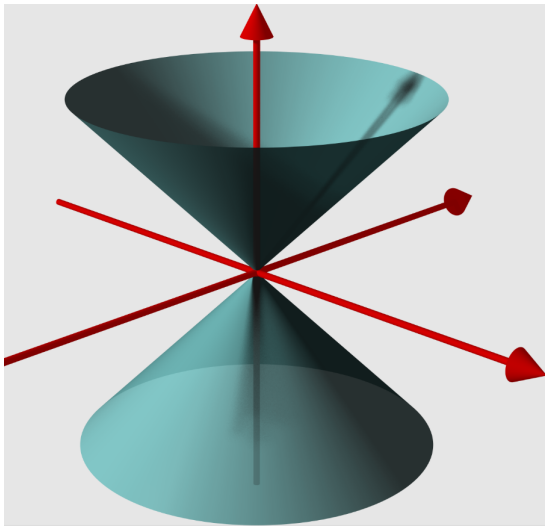


Example 3.7

Let $A = \{(x, y, z) \mid z^2 = x^2 + y^2\}$. Then A is not a regular surface. However,

$$S = A \setminus \{(0, 0, 0)\}$$

is a regular surface.



Theorem 3.4

Let $\mathcal{U}$ be an open subset of $\mathbb{R}^3$, and let $f : \mathcal{U} \to \mathbb{R}$ be a smooth function. Define

$$S = \{(x, y, z) \in \mathcal{U} \mid f(x, y, z) = 0\}.$$

If $\nabla f(x, y, z) \neq \mathbf{0}$ for all $(x, y, z) \in S$, then S is a regular surface.

A surface of the form $f(x, y, z) = 0$ is called a level surface.

Example 3.8

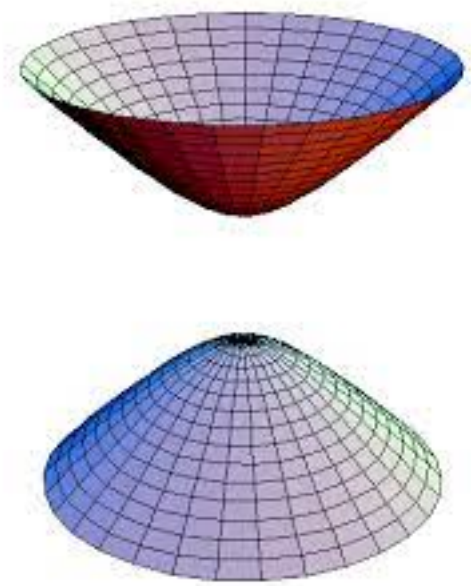
The sphere $x^2 + y^2 + z^2 = 1$ is a level surface.

Example 3.9

Show that the hyperboloid of two sheets

$$S = \{(x, y, z) \mid z^2 - x^2 - y^2 = 1\}$$

is a regular surface.



Theorem 3.5

Let S be a regular surface. For any (x_0, y_0, z_0) in S, there is an open set U in $\mathbb{R}^2$, an open set V in $\mathbb{R}^3$, and a smooth function $f : U \to \mathbb{R}$ such that $V \cap S$ has one of the following three forms.

$$\{(x, y, z) \,|\, (x, y) \in U,\ z = f(x, y)\},$$
$$\{(x, y, z) \,|\, (x, z) \in U,\ y = f(x, z)\},$$
$$\{(x, y, z) \,|\, (y, z) \in U,\ x = f(y, z)\}.$$

Definition 3.2 Transition Maps

Let S be a regular surface. Assume that $\boldsymbol{\sigma}_1 : \mathcal{O}_1 \to S \cap V_1$ and $\boldsymbol{\sigma}_2 : \mathcal{O}_2 \to S \cap V_2$ are two coordinate patches and $S \cap V_1 \cap V_2 \neq \emptyset$. Then $\mathcal{U}_1 = \boldsymbol{\sigma}_1^{-1}(V_1 \cap V_2)$ and $\mathcal{U}_2 = \boldsymbol{\sigma}_2^{-1}(V_1 \cap V_2)$ are nonempty open subsets of $\mathbb{R}^2$ that are contained in $\mathcal{O}_1$ and $\mathcal{O}_2$ respectively. The map $\boldsymbol{\Phi} : \mathcal{U}_1 \to \mathcal{U}_2$ such that

$$\boldsymbol{\sigma}_1(u, v) = (\boldsymbol{\sigma}_2 \circ \boldsymbol{\Phi})(u, v) \qquad \text{for all } (u, v) \in \mathcal{U}_1$$

is called a transition map.

Example 3.10

Let

$$\boldsymbol{\sigma}_1(u,v) = (6-2u-3v, u, v)\,, \quad \boldsymbol{\sigma}_2(u,v) = \left(u, v, \frac{6-u-2v}{3}\right).$$

They are both parametrization of the plane $x+2y+3z=6$. Find the transition map $\boldsymbol{\Phi}$ such that $\boldsymbol{\sigma}_1 = \boldsymbol{\sigma}_2 \circ \boldsymbol{\Phi}$.

Theorem 3.6

Let S be a regular surface, and let $\boldsymbol{\sigma}_1 : \mathcal{O}_1 \to S \cap V_1$ and $\boldsymbol{\sigma}_2 : \mathcal{O}_2 \to S \cap V_2$ be two coordinate patches of S. Assume that $S \cap V_1 \cap V_2 \neq \emptyset$. Let $\boldsymbol{\Phi} : \boldsymbol{\sigma}_1^{-1}(V_1 \cap V_2) \to \boldsymbol{\sigma}_2^{-1}(V_1 \cap V_2)$ be the transition map such that

$$\boldsymbol{\sigma}_1 = \boldsymbol{\sigma}_2 \circ \boldsymbol{\Phi}.$$

Then $\boldsymbol{\Phi}$ is a smooth bijective map, and for any $(u, v) \in \boldsymbol{\sigma}_1^{-1}(V_1 \cap V_2)$, $\det \mathbf{D}\boldsymbol{\Phi}(u, v) \neq 0$.

We want to define smooth functions on a surface.

Definition 3.3 Functions on Surfaces

Let S be a regular surface. A function $f : S \to \mathbb{R}$ is smooth at a point $\mathbf{x}_0 \in S$ if for any coordinate patch $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ whose image contains $\mathbf{x}_0$, $(f \circ \boldsymbol{\sigma}) : \mathcal{O} \to \mathbb{R}$ is smooth at the point $\boldsymbol{\sigma}^{-1}(\mathbf{x}_0)$.

The function $f : S \to \mathbb{R}$ is a smooth function if it is smooth at every point of S.

If $\boldsymbol{\sigma}_1 : \mathcal{O}_1 \to S \cap V_1$ and $\boldsymbol{\sigma}_2 : \mathcal{O}_2 \to S \cap V_2$ are two coordinate patches at the point $\mathbf{x}_0$, there is a smooth transition map $\boldsymbol{\Phi} : \boldsymbol{\sigma}_1^{-1}(V_1 \cap V_2) \to \boldsymbol{\sigma}_2^{-1}(V_1 \cap V_2)$ such that

$$\boldsymbol{\sigma}_1 = \boldsymbol{\sigma}_2 \circ \Phi.$$

This implies that if $(f \circ \boldsymbol{\sigma}_1) : \mathcal{O}_1 \to \mathbb{R}$ is smooth at the point $\boldsymbol{\sigma}_1^{-1}(\mathbf{x}_0)$, then $(f \circ \boldsymbol{\sigma}_2) : \mathcal{O}_2 \to \mathbb{R}$ is smooth at the point $\boldsymbol{\sigma}_2^{-1}(\mathbf{x}_0)$. Thus, to check that $f : S \to \mathbb{R}$ is smooth at $\mathbf{x}_0 \in S$, it is sufficient to check for one of the coordinate patches whose image contains $\mathbf{x}_0$.

Proposition 3.7

If W is an open subset of $\mathbb{R}^3$ that contains the regular surface S, and $f : W \to \mathbb{R}$ is a smooth function, then $f|_S : S \to \mathbb{R}$ is a smooth function on S.

Example 3.11

Let S be the sphere $x^2 + y^2 + z^2 = 4$. The function $f : S \to \mathbb{R}$, $f(x, y, z) = x + y$ is a smooth function on S.

Definition 3.4 Maps Between Surfaces

Let S_1 and S_2 be regular surfaces in $\mathbb{R}^3$. A map $f : S_1 \to S_2$ is smooth at the point $\mathbf{x}_0 \in S_1$ if whenever $\boldsymbol{\sigma}_1 : \mathcal{O}_1 \to S_1 \cap V_1$ is a coordinate patch of S_1 at $\mathbf{x}_0$, $\boldsymbol{\sigma}_2 : \mathcal{O}_2 \to S_2 \cap V_2$ is a coordinate patch of S_2 such that $f(S_1 \cap V_1) \subset S_2 \cap V_2$, the map $\boldsymbol{\sigma}_2^{-1} \circ f \circ \boldsymbol{\sigma}_1 : \mathcal{O}_1 \to \mathcal{O}_2$ is smooth at $\boldsymbol{\sigma}_1^{-1}(\mathbf{x}_0)$.
We say that $f : S_1 \to S_2$ is a smooth map if it is smooth at each point of S_1.

Again, it is sufficient to check for one coordinate patch for each point.

Proposition 3.8

Let S_1 and S_2 be regular surfaces, and let $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be a smooth function such that $\mathbf{F}(S_1) \subset S_2$. Then the map $f : S_1 \to S_2$ defined by $f(\mathbf{x}) = \mathbf{F}(\mathbf{x})$ for all $\mathbf{x} \in S_1$ is a smooth map.

Example 3.12

Let

$$S_1 = \left\{(x, y, z) \,|\, x^2 + y^2 + z^2 = 1\right\},$$

$$S_2 = \left\{(x, y, z) \,|\, 4x^2 + 9y^2 + 36z^2 = 36\right\}.$$

Define the map $f : S_1 \to \mathbb{R}^3$ by

$$f(x, y, z) = (3x, 2y, z).$$

Then $f(S_1) \subset S_2$ and $f : S_1 \to S_2$ is a smooth map.

Example 3.13

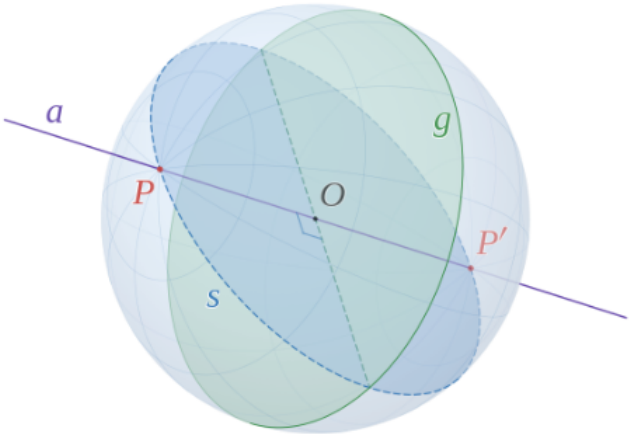
Let S be the sphere

$$S = \left\{(x, y, z) \,|\, x^2 + y^2 + z^2 = 1\right\},$$

and let $f : S \to S$ be the function

$$f(x, y, z) = (-x, -y, -z).$$

Then $f : S \to S$ is a smooth map.



Definition 3.5 Diffeomorphism

Let S_1 and S_2 be two regular surfaces. We say that $f : S_1 \to S_2$ is a diffeomorphism if $f : S_1 \to S_2$ is a smooth bijective map such that $f^{-1} : S_2 \to S_1$ is also smooth. If there is a diffeomorphism $f : S_1 \to S_2$ between S_1 and S_2, then we say that S_1 and S_2 are diffeomorphic, or isomorphic.

Example 3.14

Let

$$S_1 = \left\{(x, y, z) \,|\, x^2 + y^2 + z^2 = 1\right\},$$

$$S_2 = \left\{(x, y, z) \,|\, 4x^2 + 9y^2 + 36z^2 = 36\right\}.$$

The map $f : S_1 \to S_2$,

$$f(x, y, z) = (3x, 2y, z),$$

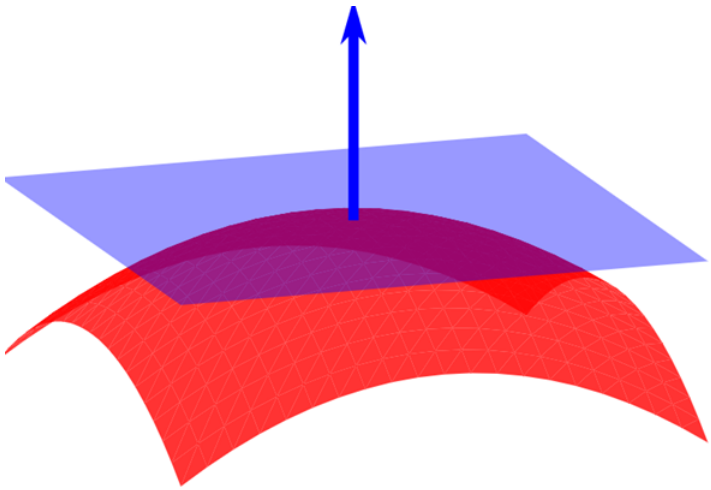
is a diffeomorphism.

3.2 Tangents and Normals

Definition 3.6 Tangent Space

Let S be a regular surface and let $\mathbf{x}_0$ be a point on S. A tangent vector to the surface S at $\mathbf{x}_0$ is the tangent vector to a regular curve on S that passes through $\mathbf{x}_0$.

The tangent space of S at $\mathbf{x}_0$, denoted by $T_{\mathbf{x}_0}S$, is the set of all the tangent vectors to the surface S at $\mathbf{x}_0$.



Proposition 3.9

Let S be a regular surface and let $\mathbf{x}_0$ be a point on S. If $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ is a coordinate patch such that $\boldsymbol{\sigma}(u_0, v_0) = \mathbf{x}_0$, define the parameter curves as

$$\boldsymbol{\gamma}_1(t) = \boldsymbol{\sigma}(t, v_0) \quad \text{and} \quad \boldsymbol{\gamma}_2(t) = \boldsymbol{\sigma}(u_0, t).$$

Then

$$\boldsymbol{\gamma}_1'(u_0) = \boldsymbol{\sigma}_u(u_0, v_0) \quad \text{and} \quad \boldsymbol{\gamma}_2'(v_0) = \boldsymbol{\sigma}_v(u_0, v_0)$$

Hence, $\boldsymbol{\sigma}_u(u_0, v_0)$ and $\boldsymbol{\sigma}_v(u_0, v_0)$ are tangent vectors to the surface S at $\mathbf{x}_0$.

Example 3.15

Consider the plane $x + 2y + 3z = 6$ and the point $\mathbf{x}_0 = (x_0, y_0, z_0) = (1, 1, 1)$ on the plane. If $\boldsymbol{\gamma} : (a, b) \to \mathbb{R}^3$ is a regular curve on the plane with $\boldsymbol{\gamma}(t_0) = \mathbf{x}_0$, then

$$\gamma_1(t) + 2\gamma_2(t) + 3\gamma_3(t) = 6.$$

Hence,

$$\gamma_1'(t_0) + 2\gamma_2'(t_0) + 3\gamma_3'(t_0) = 0.$$

This shows that the tangent vector $\boldsymbol{\gamma}'(t_0)$ is perpendicular to $\mathbf{a} = (1, 2, 3)$, the normal vector to the plane.

The tangent space W at $\mathbf{x}_0$ is the 2-dimensional subspace of $\mathbb{R}^3$ which contains all the vectors that are perpendicular to $\mathbf{a}$.

Lemma 3.10

Let $\mathcal{O}$ and $\mathcal{U}$ be open subsets of $\mathbb{R}^2$, and let $\mathbf{\Phi} : \mathcal{U} \to \mathcal{O}$ be a diffeomorphism. Assume that $\mathbf{b}$ is a point in $\mathcal{U}$ and $\mathbf{\Phi}(\mathbf{b}) = \mathbf{a}$. Let $\boldsymbol{\sigma} : \mathcal{O} \to \mathbb{R}^3$ be a smooth map such that $\boldsymbol{\sigma}_u(\mathbf{a})$ and $\boldsymbol{\sigma}_v(\mathbf{a})$ are linearly independent. Define the map $\widetilde{\boldsymbol{\sigma}} : \mathcal{U} \to \mathbb{R}^3$ by

$$\widetilde{\boldsymbol{\sigma}} = \boldsymbol{\sigma} \circ \mathbf{\Phi}.$$

Then

$$\widetilde{\boldsymbol{\sigma}}_u(\mathbf{b}) \times \widetilde{\boldsymbol{\sigma}}_v(\mathbf{b}) = \det(\mathbf{D}\mathbf{\Phi}(\mathbf{b}))\left(\boldsymbol{\sigma}_u(\mathbf{a}) \times \boldsymbol{\sigma}_v(\mathbf{a})\right).$$

Therefore, $\widetilde{\boldsymbol{\sigma}}_u(\mathbf{b})$ and $\widetilde{\boldsymbol{\sigma}}_v(\mathbf{b})$ are linearly independent. Moreover, the subspace of $\mathbb{R}^3$ spanned by $\widetilde{\boldsymbol{\sigma}}_u(\mathbf{b})$ and $\widetilde{\boldsymbol{\sigma}}_v(\mathbf{b})$ is the same as the subspace of $\mathbb{R}^3$ spanned by $\boldsymbol{\sigma}_u(\mathbf{a})$ and $\boldsymbol{\sigma}_v(\mathbf{a})$.

Lemma 3.11

Let S be a regular surface. Assume that $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ is a coordinate patch of S, $\boldsymbol{\gamma} : I \to S \cap V$ is a curve on $S \cap V$. Then there exists a smooth map $\boldsymbol{\alpha} : I \to \mathcal{O}$ such that

$$\boldsymbol{\gamma} = \boldsymbol{\sigma} \circ \boldsymbol{\alpha}.$$

Theorem 3.12

Let S be a regular surface and let $\mathbf{x}_0$ be a point on S. If $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ is a coordinate patch such that $\boldsymbol{\sigma}(\mathbf{a}) = \mathbf{x}_0$, then the tangent space of S at $\mathbf{x}_0$ is the two-dimensional subspace of $\mathbb{R}^3$ that is spanned by $\boldsymbol{\sigma}_u(\mathbf{a})$ and $\boldsymbol{\sigma}_v(\mathbf{a})$.

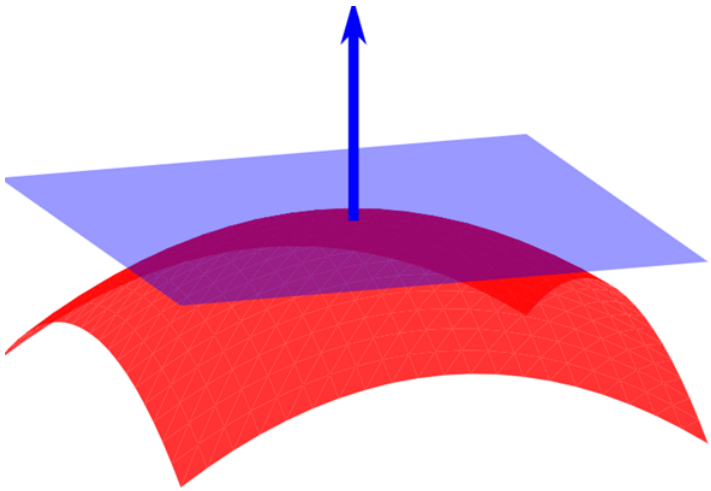
Lemma 3.10 implies that the tangent space is independent of parametrization.

Definition 3.7 Tangent Planes

Let S be a regular surface and let $\mathbf{x}_0$ be a point on S. If $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ is a coordinate patch such that $\boldsymbol{\sigma}(\mathbf{a}) = \mathbf{x}_0$, then the tangent plane of S at $\mathbf{x}_0$ is the plane with parametric equation

$$\mathbf{x} = \mathbf{x}_0 + t_1 \boldsymbol{\sigma}_u(\mathbf{a}) + t_2 \boldsymbol{\sigma}_v(\mathbf{a}), \qquad (t_1, t_2) \in \mathbb{R}^2.$$

A normal vector to the plane is given by $\boldsymbol{\sigma}_u(\mathbf{a}) \times \boldsymbol{\sigma}_v(\mathbf{a})$.



Theorem 3.13

Let $\mathcal{O}$ be an open subset of $\mathbb{R}^2$, and let $f : \mathcal{O} \to \mathbb{R}$ be a smooth function. Then the graph of f, which is the subset of $\mathbb{R}^3$ given by

$$G_f = \{(x, y, z) \,|\, (x, y) \in \mathcal{O}, z = f(x, y)\},$$

is a regular surface. The tangent plane at the point $(x_0, y_0, f(x_0, y_0))$ is

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Theorem 3.14

Let $\mathcal{U}$ be an open subset of $\mathbb{R}^3$, and let $f : \mathcal{U} \to \mathbb{R}$ be a smooth function. Define

$$S = \{(x, y, z) \in \mathcal{U} \mid f(x, y, z) = 0\}.$$

If $\nabla f(x_0, y_0, z_0) \neq \mathbf{0}$ for all $(x_0, y_0, z_0) \in S$, then S is a regular surface. The equation of the tangent plane at the point (x_0, y_0, z_0) is

$$f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) = 0.$$

Example 3.16

Let a, b, c be positive numbers, and let S be the ellipsoid

$$S = \left\{ (x, y, z) \;\middle|\; \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \right\}.$$

If (x_0, y_0, z_0) is a point on S, find the equation of the tangent plane at (x_0, y_0, z_0).

Definition 3.8 Standard Unit Normal Vector

Let S be a regular surface and let $\mathbf{x}_0$ be a point on S. If $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ is a coordinate patch such that $\boldsymbol{\sigma}(\mathbf{a}) = \mathbf{x}_0$, then the vector

$$\mathbf{N}_{\boldsymbol{\sigma}} = \frac{\boldsymbol{\sigma}_u(\mathbf{a}) \times \boldsymbol{\sigma}_v(\mathbf{a})}{\|\boldsymbol{\sigma}_u(\mathbf{a}) \times \boldsymbol{\sigma}_v(\mathbf{a})\|}$$

is the standard unit normal vector of the coordinate patch $\boldsymbol{\sigma}$ at the point $\mathbf{x}_0$.

The standard unit normal vector depends on parametrizations up to a sign.

Example 3.17

If S is the plane $x+2y+3z=6$, then the tangent plane at any point $\mathbf{x}_0$ on S is the plane itself. A normal vector to the surface is $\mathbf{a}=(1,2,3)$.

If we parametrize S by $\boldsymbol{\sigma}:\mathbb{R}^2\to S$,

$$\boldsymbol{\sigma}(u,v)=(6-2u-3v,u,v)\,,$$

then

$$\boldsymbol{\sigma}_u=(-2,1,0),\quad \boldsymbol{\sigma}_v=(-3,0,1),\quad \boldsymbol{\sigma}_u\times\boldsymbol{\sigma}_v=(1,2,3),$$

and so

$$\mathbf{N}_{\boldsymbol{\sigma}}=\frac{1}{\sqrt{14}}(1,2,3).$$

If we parametrize S by $\widetilde{\boldsymbol{\sigma}}:\mathbb{R}^2\to S$,

$$\widetilde{\boldsymbol{\sigma}}(u,v)=\left(u,\frac{6-u-3v}{2},v\right),$$

then

$$\widetilde{\boldsymbol{\sigma}}_u=\left(1,-\frac{1}{2},0\right),\quad \widetilde{\boldsymbol{\sigma}}_v=\left(0,-\frac{3}{2},1\right),\quad \widetilde{\boldsymbol{\sigma}}_u\times\widetilde{\boldsymbol{\sigma}}_v=-\frac{1}{2}(1,2,3),$$

and so

$$\mathbf{N}_{\widetilde{\boldsymbol{\sigma}}}=-\frac{1}{\sqrt{14}}(1,2,3).$$

In fact, the transition map $\boldsymbol{\Phi}:\mathbb{R}^2\to\mathbb{R}^2$ such that $\widetilde{\boldsymbol{\sigma}}=\boldsymbol{\sigma}\circ\boldsymbol{\Phi}$ is given by

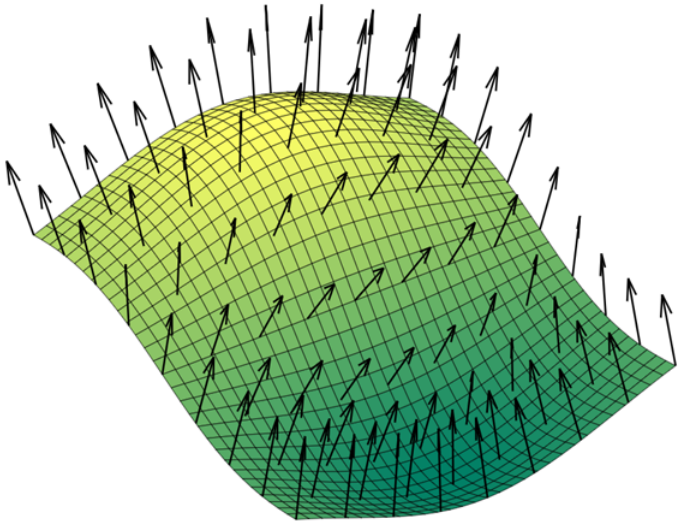
$$\boldsymbol{\Phi}(u,v)=\left(\frac{6-u-3v}{2},v\right).$$

Therefore,

$$\mathbf{D}\boldsymbol{\Phi}(u,v)=\begin{bmatrix}-\frac{1}{2} & -\frac{3}{2}\\ 0 & 1\end{bmatrix},\qquad \det\mathbf{D}\boldsymbol{\Phi}(u,v)=-\frac{1}{2}.$$

Definition 3.9 Orientability

Let S be a regular surface. We say that S is orientable if there is a an atlas $\mathcal{A}$ of coordinate patches so that if $\mathbf{\Phi}$ is the transition map between two overlapping patches in $\mathcal{A}$, then $\det(\mathbf{D\Phi}) > 0$.



Proposition 3.15

Let S be a regular surface. Then the following statements are equivalent.

(a) S is an orientable surface, in the sense that there is an atlas $\mathcal{A}$ of coordinate patches so that $\det(\mathbf{D}\mathbf{\Phi}) > 0$ for the transition map $\mathbf{\Phi}$ between two any two overlapping patches in $\mathcal{A}$.

(b) There is a continuous choice of unit normal vectors on S.

Example 3.18

Any plane in $\mathbb{R}^3$ is orientable.

Example 3.19

An open subset of an orientable surface is orientable.

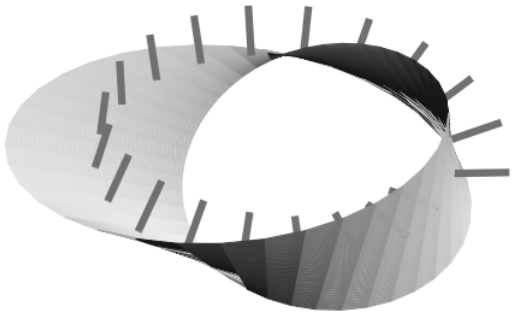
Example 3.20

A sphere $x^2 + y^2 + z^2 = 1$ is orientable.

Example 3.21

The Möbius band is not orientable.

The Möbius band is a regular surface S obtained by rotating a straightline segment L around its midpoint $\mathbf{x}_0$ at the same time as $\mathbf{x}_0$ moves around a circle C, in such a way that when $\mathbf{x}_0$ moves once around C, L rotates $180°$ around $\mathbf{x}_0$.



Take C to be the circle $x^2 + y^2 = 1$ on the $z = 0$ plane, and let L be the line segment $(1, 0, t)$, $-\frac{1}{2} < t < \frac{1}{2}$), so that $\mathbf{x}_0 = (1, 0, 0)$. A point $\mathbf{x}$ on the Möbius band can be described by two parameters t and θ, with

$$\mathbf{x} = (x(t,\theta), y(t,\theta), z(t,\theta)) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 - t\sin\frac{\theta}{2} \\ 0 \\ t\cos\frac{\theta}{2} \end{bmatrix},$$

where

$$-\frac{1}{2} < t < \frac{1}{2}, \quad 0 \le \theta < 2\pi.$$

A straightforward computation gives

$$\mathbf{x}_t(t,\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -\sin\frac{\theta}{2} \\ 0 \\ \cos\frac{\theta}{2} \end{bmatrix};$$

$$\mathbf{x}_\theta(t,\theta) = \begin{bmatrix} -\sin\theta & -\cos\theta & 0 \\ \cos\theta & -\sin\theta & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 - t\sin\frac{\theta}{2} \\ 0 \\ t\cos\frac{\theta}{2} \end{bmatrix} - \frac{t}{2} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\frac{\theta}{2} \\ 0 \\ \sin\frac{\theta}{2} \end{bmatrix}.$$

An atlas of the surface is given by $\{\boldsymbol{\sigma}_1, \boldsymbol{\sigma}_2\}$, where $\boldsymbol{\sigma}_1 : (-\frac{1}{2}, \frac{1}{2}) \times (0, 2\pi) \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}_1(t, \theta) = (x(t, \theta), y(t, \theta), z(t, \theta))\,;$$

and $\boldsymbol{\sigma}_2 : (-\frac{1}{2}, \frac{1}{2}) \times (-\pi, \pi) \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}_2(t, \theta) = (x(t, \theta), y(t, \theta), z(t, \theta))\,.$$

The standard unit normal vector of the coordinate patch $\boldsymbol{\sigma}_1$ is

$$\mathbf{N}_{\boldsymbol{\sigma}_1}(t, \theta) = \frac{(\boldsymbol{\sigma}_1)_t(t, \theta) \times (\boldsymbol{\sigma}_1)_\theta(t, \theta)}{\|(\boldsymbol{\sigma}_1)_t(t, \theta) \times (\boldsymbol{\sigma}_1)_\theta(t, \theta)\|}.$$

It varies continuously on $S \setminus L$. As $\theta \to 0^+$,

$$\mathbf{x}_t(0, \theta) \to (0, 0, 1), \qquad \mathbf{x}_\theta(0, \theta) \to (0, 1, 0)$$

and so

$$\mathbf{N}_{\boldsymbol{\sigma}_1}(0, \theta) \to (-1, 0, 0).$$

As $\theta \to 2\pi^-$,

$$\mathbf{x}_t(0, \theta) \to (0, 0, -1), \qquad \mathbf{x}_\theta(0, \theta) \to (0, 1, 0),$$

and so

$$\mathbf{N}_{\boldsymbol{\sigma}_1}(0, \theta) \to (1, 0, 0).$$

This shows that it is impossible to have a consistence choice of unit normal vectors that varies continuously on S. Hence, S is not orientable.

3.3 Derivatives

Definition 3.10 Derivatives of Smooth Maps Between Surfaces

Let S_1 and S_2 be regular surfaces, and let $f : S_1 \to S_2$ be a smooth map from S_1 to S_2. Given a point $\mathbf{x}_0$ on S_1, the derivative of f at $\mathbf{x}_0$ is the map $D_{\mathbf{x}_0} f : T_{\mathbf{x}_0} S_1 \to T_{f(\mathbf{x}_0)} S_2$, such that if $\mathbf{w}$ is the tangent vector to the curve $\boldsymbol{\gamma} : I \to S_1$ at the point $\mathbf{x}_0$, then $D_{\mathbf{x}_0} f(\mathbf{w})$ is the tangent vector to the curve $(f \circ \boldsymbol{\gamma}) : I \to S_2$ at the point $f(\mathbf{x}_0)$.

Example 3.22

Let

$$S_1 = \left\{(x, y, z) \,|\, x^2 + y^2 + z^2 = 3\right\},$$

$$S_2 = \left\{(x, y, z) \,|\, 4x^2 + y^2 + 9z^2 = 108\right\},$$

and let $f : S_1 \to S_2$ be the map

$$f(x, y, z) = (3x, 6y, 2z).$$

Find bases of $T_{\mathbf{x}_0}S_1$ and $T_{f(\mathbf{x}_0)}S_2$, and compute the derivative map $D_{\mathbf{x}_0}f : T_{\mathbf{x}_0}S_1 \to T_{f(\mathbf{x}_0)}S_2$ at the point $\mathbf{x}_0 = (1, 1, 1)$.

Theorem 3.16

Let S_1 and S_2 be regular surfaces, and let $f : S_1 \to S_2$ be a smooth function from S_1 to S_2. At any point $\mathbf{x}_0$ in S_1, the derivative map $D_{\mathbf{x}_0} f : T_{\mathbf{x}_0} S_1 \to T_{f(\mathbf{x}_0)} S_2$ is a linear transformation.

Theorem 3.17 Chain rule

Let S_1, S_2 and S_3 be regular surfaces. If $f : S_1 \to S_2$ and $g : S_2 \to S_3$ are smooth maps, $\mathbf{x}_0$ is a point in S_1, then

$$D_{\mathbf{x}_0}(g \circ f) = (D_{f(\mathbf{x}_0)}g) \circ (D_{\mathbf{x}_0}f).$$

Corollary 3.18

Let S_1 and S_2 are regular surfaces, and let $f : S_1 \to S_2$ be a diffeomorphism between S_1 and S_2. For any $\mathbf{x}_0 \in S_1$, the derivative map $D_{\mathbf{x}_0} f : T_{\mathbf{x}_0} S_1 \to T_{f(\mathbf{x}_0)} S_2$ is an invertible linear transformation.

Theorem 3.19

Let S_1 and S_2 be regular surfaces in $\mathbb{R}^3$, and let $f : S_1 \to S_2$ be a diffeomorphism. Assume that $\boldsymbol{\sigma}_1 : \mathcal{O} \to S_1 \cap V_1$ is a coordinate patch of S_1. Then there is an open subset V_2 of $\mathbb{R}^3$ so that $\boldsymbol{\sigma}_2 = (f \circ \boldsymbol{\sigma}_1) : \mathcal{O} \to S_2 \cap V_2$ is a coordinate patch of S_2. Given $\mathbf{a} \in \mathcal{O}$, let $\mathbf{x}_0 = \boldsymbol{\sigma}_1(\mathbf{a})$. Then $f(\mathbf{x}_0) = \boldsymbol{\sigma}_2(\mathbf{a})$, and

$$D_{\mathbf{x}_0} f(\boldsymbol{\sigma}_{1,u}(\mathbf{a})) = \boldsymbol{\sigma}_{2,u}(\mathbf{a}), \qquad D_{\mathbf{x}_0} f(\boldsymbol{\sigma}_{1,v}(\mathbf{a})) = \boldsymbol{\sigma}_{2,v}(\mathbf{a}).$$

Definition 3.11 Local Diffeomorphism

Let S_1 and S_2 be regular surfaces, and let $f : S_1 \to S_2$ be a smooth map from S_1 to S_2. We say that $f : S_1 \to S_2$ is a local diffeomorphism if for each $\mathbf{x}_0$ in S_1, there is an open subset U of S_1 containing $\mathbf{x}_0$ such that $V = f(U)$ is an open subset of S_2, and $f|_U : U \to V$ is a diffeomorphism.

Example 3.23

If $f : S_1 \to S_2$ is a diffeomorphism, then it is a local diffeomorphism.

Proposition 3.20

Let S_1 and S_2 be regular surfaces, and let $f : S_1 \to S_2$ be a local diffeomorphism. Then $f : S_1 \to S_2$ is a diffeomorphism if and only if it is bijective.

Theorem 3.21 Characterization of Local Diffeomorphisms

Let S_1 and S_2 be regular surfaces, and let $f : S_1 \to S_2$ be a smooth map from S_1 to S_2. Then $f : S_1 \to S_2$ is a local diffeomorphism if and only if for each $\mathbf{x}_0$ in S_1, the derivative map $D_{\mathbf{x}_0} f : T_{\mathbf{x}_0} S_1 \to T_{f(\mathbf{x}_0)} S_2$ is an invertible linear transformation.

Example 3.24

Let S_1 and S_2 be the surfaces

$$S_1 = \{(x, y, z) \mid y = 0\}, \qquad S_2 = \left\{(x, y, z) \mid x^2 + y^2 = 1\right\}.$$

Then the map $f : S_1 \to S_2$,

$$f(x, 0, z) = (\cos x, \sin x, z)$$

is a local diffeomorphism, but it is not a diffeomorphism.

Corollary 3.22 Characterization of Diffeomorphisms

Let S_1 and S_2 be regular surfaces, and let $f : S_1 \to S_2$ be a bijective smooth map from S_1 to S_2. Then $f : S_1 \to S_2$ is a diffeomorphism if and only if for each $\mathbf{x}_0$ in S_1, the derivative map $D_{\mathbf{x}_0} f : T_{\mathbf{x}_0} S_1 \to T_{f(\mathbf{x}_0)} S_2$ is an invertible linear transformation.

3.4 Special Types of Surfaces

Definition 3.12 Quadric Surfaces

A quadric is defined by a quadratic function

$$f(x, y, z) = ax^2 + by^2 + cz^2 + 2dxy + 2eyz + 2fxz + gx + hy + kz + m.$$

It is the set of points (x, y, z) that satisfy the equation $f(x, y, z) = 0$. Hence, it can be written as

$$\mathbf{x}^T \mathbf{A}\mathbf{x} + \mathbf{q}^T \mathbf{x} + m = 0,$$

where

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a & d & f \\ d & b & e \\ f & e & c \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} g \\ h \\ k \end{bmatrix}.$$

A quadric is not necessarily a regular surface. For example,

(a) $x^2 + y^2 + z^2 = 0$ is a single point $(x, y, z) = (0, 0, 0)$.

(b) $x^2 + y^2 = 0$ is the z-axis.

Using the theory of symmetric matrices, there is an orthogonal matrix $\mathbf{P}$ and a diagonal matrix

$$\mathbf{D} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix},$$

so that

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T.$$

If λ_1, λ_2 and λ_3 are all nonzero, then we can do translations to eliminate the degree one terms. In this case, the quadric has one of the following forms.

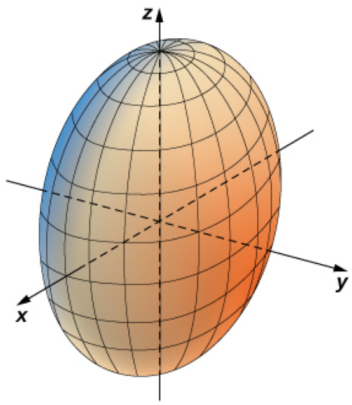
(a) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$, an ellipsoid.

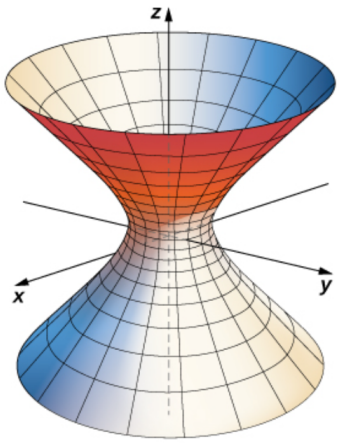
(b) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$, a hyperboloid of one sheet.

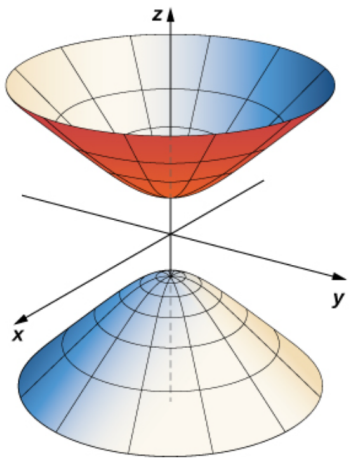
(c) $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$, a hyperboloid of two sheets.

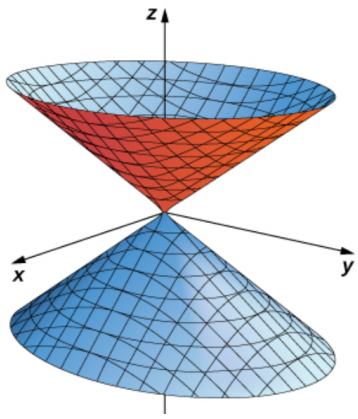
(d) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = \dfrac{z^2}{c^2}$, a quadric cone.

(e) A single point $(0, 0, 0)$









If exactly one of the λ_i is zero, then the quadric has one of the following forms.

(f) $z = \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2}$, an elliptic paraboloid.

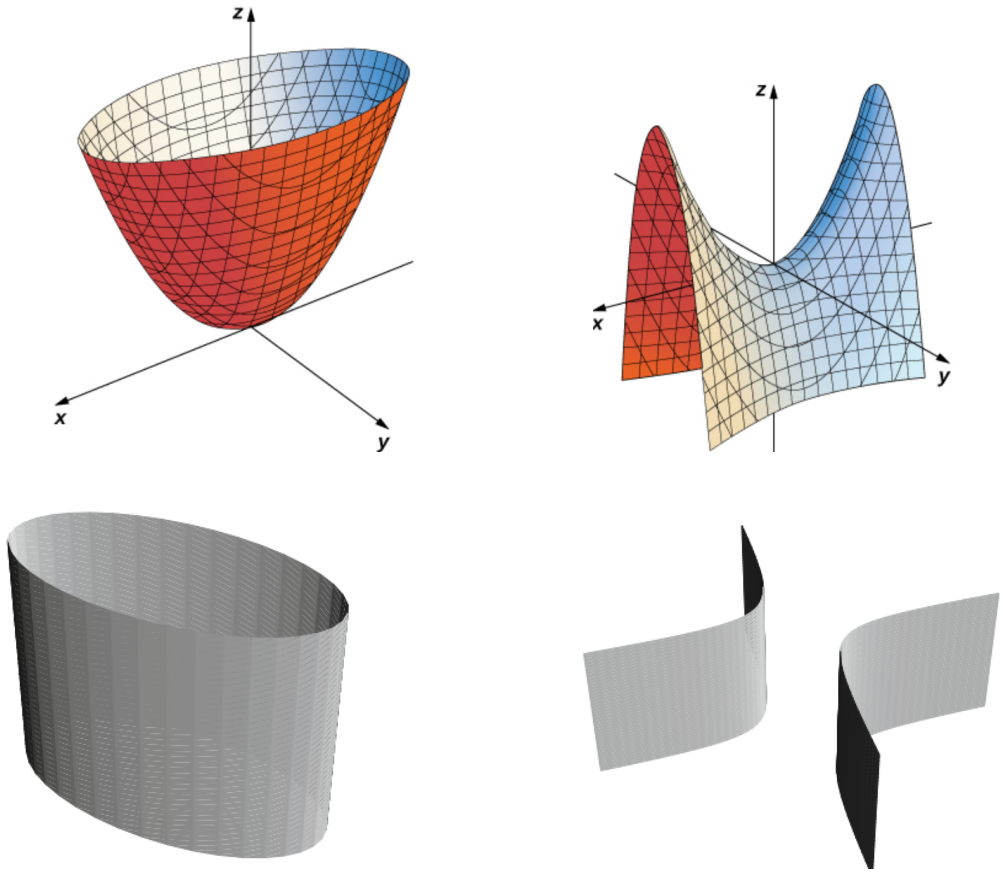
(g) $z = \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2}$, a hyperbolic paraboloid.

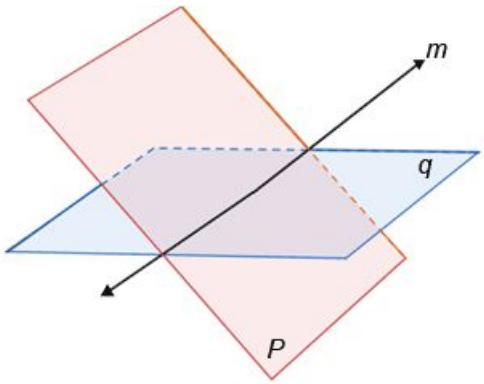
(h) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, an elliptic cylinder.

(i) $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$, a hyperbolic cylinder.

(j) $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 0$, two intersecting planes.

(k) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 0$, a line.



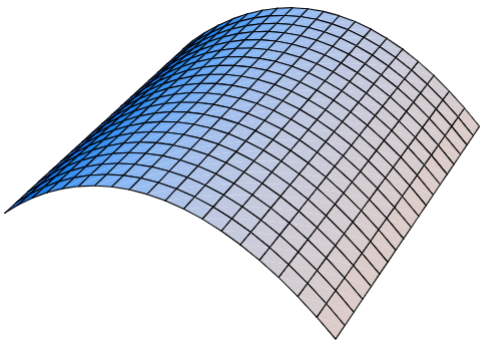


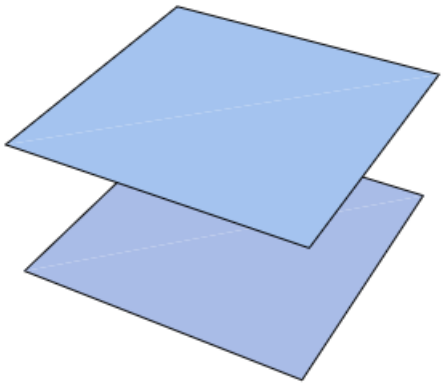
If two of the λ_i's are zero, then the quadric has one of the following forms.

(l) $y = \dfrac{x^2}{a^2}$, a parabolic cylinder.

(m) $\dfrac{x^2}{a^2} = 1$, two parallel planes.

(n) $x = 0$, one plane.





Theorem 3.23

A quadric is a regular surface if and only if it has one of the following standard forms.

(a) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$, an ellipsoid.

(b) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$, a hyperboloid of one sheet.

(c) $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$, a hyperboloid of two sheets.

(d) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = \dfrac{z^2}{c^2}$, a quadric cone with the point $(0, 0, 0)$ removed.

(e) $z = \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2}$, an elliptic paraboloid.

(f) $z = \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2}$, a hyperbolic paraboloid.

(g) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, an elliptic cylinder.

(h) $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$, a hyperbolic cylinder.

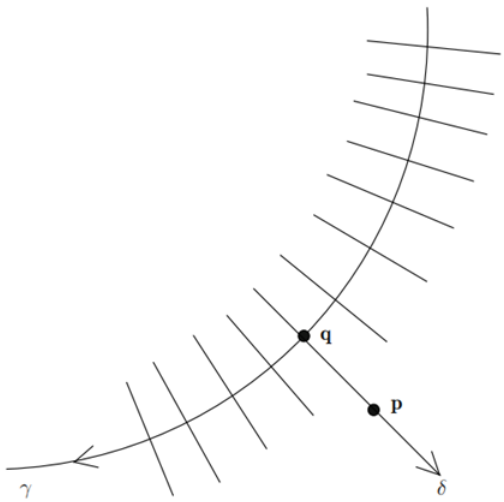
(i) $y = \dfrac{x^2}{a^2}$, a parabolic cylinder.

(j) $\dfrac{x^2}{a^2} = 1$, two parallel planes.

(k) $x = 0$, one plane.

Definition 3.13 Ruled Surfaces

A ruled surface is a surface which is a union of straightlines. The straightlines are called the rulings or the generators of the surface.

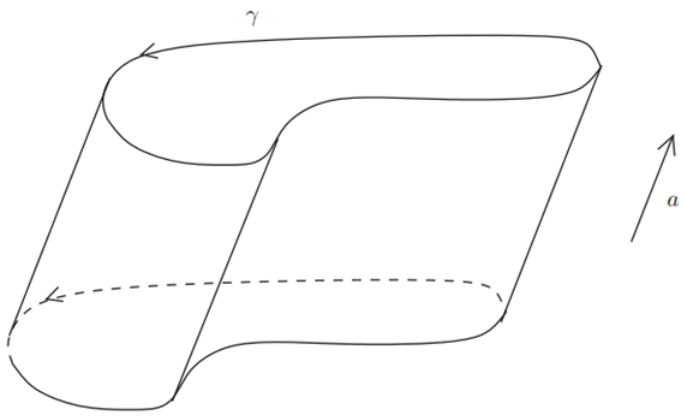


Definition 3.14 Generalized Cylinders

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a simple regular curve, and let $\mathbf{w}$ be a vector in $\mathbb{R}^3$ that is not parallel to $\boldsymbol{\gamma}'(t)$ for all $t \in I$. Then $\boldsymbol{\sigma} : I \times \mathbb{R} \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}(t, v) = \boldsymbol{\gamma}(t) + v\mathbf{w}$$

is a ruled surface, called a generalized cylinder. In this case, all the rulings are parallel.

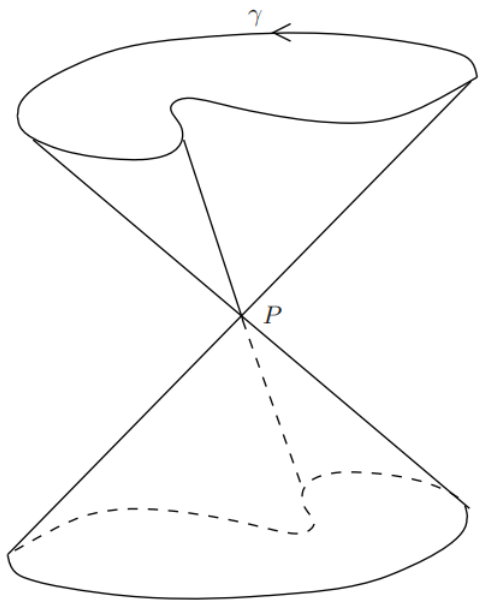


Definition 3.15 Generalized Cones

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a simple regular curve, and let $\mathbf{x}_0$ be a fixed point in $\mathbb{R}^3$ that does not lie on the curve $\boldsymbol{\gamma}$. Then $\boldsymbol{\sigma} : I \times (\mathbb{R} \setminus \{0\}) \to \mathbb{R}^3$,

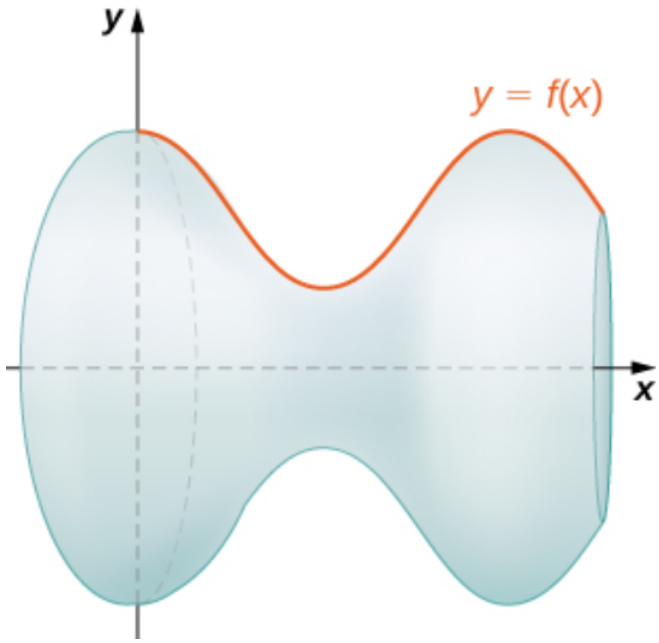
$$\boldsymbol{\sigma}(t, v) = v\boldsymbol{\gamma}(t) + (1 - v)\mathbf{x}_0$$

is a ruled surface, called a generalized cone, if for any $t \in I$, the vector $\boldsymbol{\gamma}(t) - \mathbf{x}_0$ is not parallel to the tangent vector $\boldsymbol{\gamma}'(t)$. In this case, all the rulings pass throught the point $\mathbf{x}_0$.



Definition 3.16 Surface of Revolutions

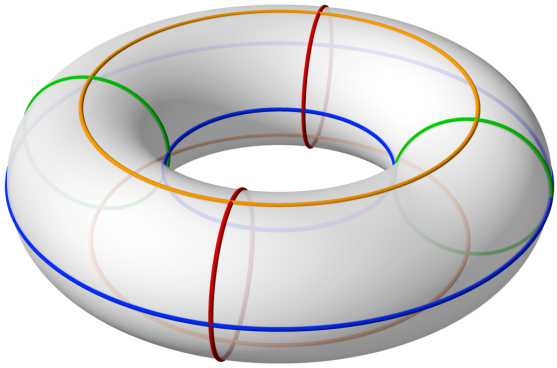
A surface of revolution is a surface obtained by rotating a plane curve, called the profile curve, around a straightline in the plane. The circles obtained by rotating a fixed point on the profile curve around the axis of rotation are called the parallels of the surface. The curves on the surface obtained by rotating the profile curve through a fixed angle are called its meridians.



Example 3.25

A sphere $x^2 + y^2 + z^2 = 1$ is a surface of revolution.

A torus is a surface of revolution obtained by rotating a circle around a straightline that lies on the plane of the circle, but does not intersect the circle.



Without loss of generality, assume that b and r are positive numbers with $0 < r < b$, and the circle is given by

$$x^2 + (y - b)^2 = r^2, \quad z = 0.$$

The circle can be parametrized by $\boldsymbol{\gamma} : [0, 2\pi] \to \mathbb{R}^3$,

$$\boldsymbol{\gamma}(u) = (r \cos u, r \sin u + b, 0).$$

Thus, every point on the torus can be written as

$$\mathbf{x}(u, v) = (r \cos u, (r \sin u + b) \cos v, (r \sin u + b) \sin v)\,,$$

where

$$0 \leq u < 2\pi, \qquad 0 \leq v < 2\pi.$$

3.5 Compact Surfaces

A subset of $\mathbb{R}^3$ is said to be compact if it is closed and bounded.

Definition 3.17 Compact Surface

A surface is compact if it is a closed and bounded subset of $\mathbb{R}^3$.

Example 3.26

A plane is not compact.

Example 3.27

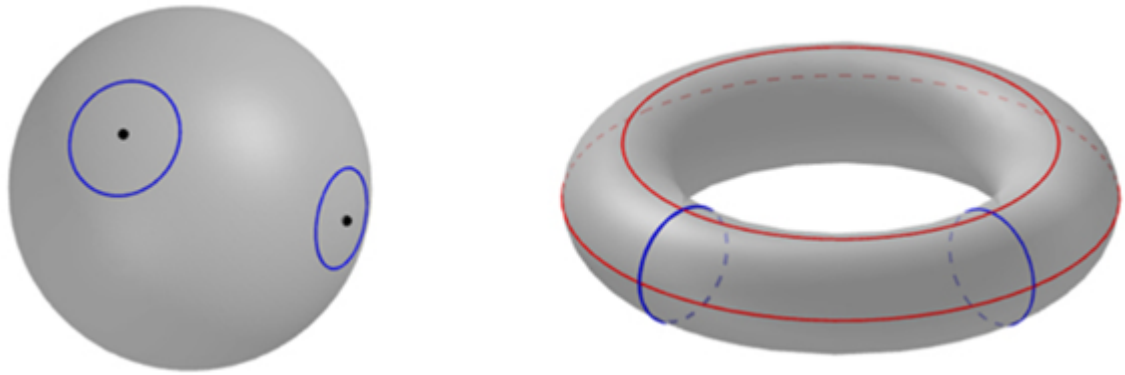
An ellipsoid is a compact surface.

Recall that if $f : \mathbb{R}^3$ is a continuous function, then the set

$$\{(x, y, z) \mid f(x, y, z) = 0\}$$

is a closed subset of $\mathbb{R}^3$.

Hence, a bounded level surface is a compact surface.



Notice that any ellipsoid is diffeomorphic to a sphere.

The sphere has no holes, while a torus has one hole.

There are compact surfaces with n holes, where n is any positive integer.

Theorem 3.24 Classification of Compact Surfaces

A compact surfaces is determined, up to diffeomorphisms, by its genus g, which is the number of holes of the surface. Every compact surface is orientable.